

Discussion of
**"Housing Market Freezes, Deleveraging, and
Aggregate Demand"**

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HOUSING, HOUSING CREDIT AND THE MACROECONOMY
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This Paper

- Very preliminary, but **very ambitious!**
- A dimension of the housing / mortgage market **less explored** within the macro literature: **liquidity**.
 - time-to-sell a house
 - assets generated from mortgage origination and re-packaging
- There seems to be a clear business cycle fluctuation of both.
- What they do / propose:
 - 1 A theory for the transmission of "**liquidity shocks**", built into a rich HANK-type model.
 - 2 Smart way to make tractable an otherwise (theoretically and computationally) very complex problem.

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Key Mechanism

- **Partial equilibrium:** a household's problem represented by the value function

$$\bar{V}(b, k, \chi; \nu, \Theta) = V_n + E\left(\max\left\{0, \nu(V_a - V_n) - \kappa\right\}\right)$$

which implies **threshold** participation strategies.

- **General equilibrium:**
 - incomplete markets
 - sticky prices
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Main Comments

Two model features which I have been thinking about regarding their mapping to the data:

- ① Is housing, in reality (i.e. data), a good match to the illiquid asset in the model?
- ② Modelling liquidity:
 - trading probability **exogenous**
 - supply of liquid assets **endogenous**

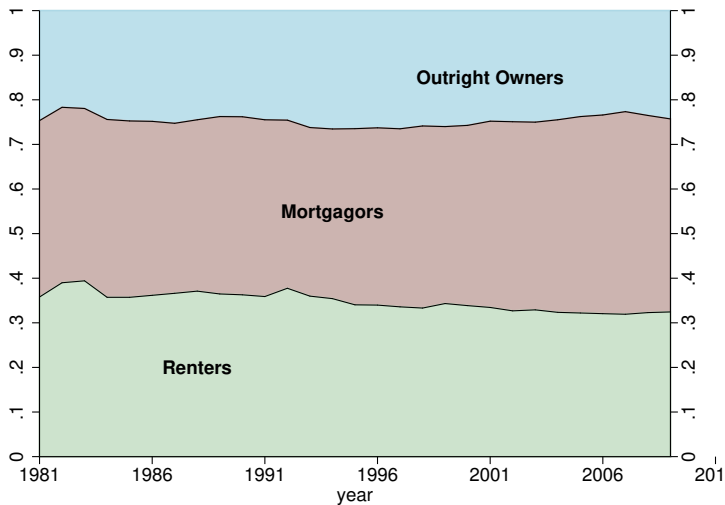
1. Housing tenure and Wealth in the US

Table: Households' (net) Financial and Housing Wealth

	Median	[p25 , p75]
Net Financial wealth		
Outright owners	5,580	[605 , 56,455]
Mortgagors	1500	[-95 , 18,705]
Renters	49	[-545 , 2,005]
Net Housing wealth		
Outright owners	165,000	[76,000 , 265,000]
Mortgagors	79,500	[32,000 , 168,000]
Renters	0	0

Constructed using data from the SCF 2004 wave

Figure: Housing tenure shares: US 1980-2008



Constructed using data from the CEX 1981-2011

Liquidity-elasticity of housing capital demand

Recall the household's problem

$$\bar{V}(b, k, \chi; \nu, \Theta) = V_n + E\left(\max\left\{0, \nu(V_a - V_n) - \kappa\right\}\right)$$

- Paper claims housing liquidity shock affects **all** households
- But, plausibly, the three housing tenure groups are affected differently.
- Households **"at the margin"**: who are they?

2. Modelling Liquidity

- Interesting feedback between mortgage finance and supply of liquidity
- The probability of trading, conditional on participating,

$$\nu_t = \frac{e^{\nu_t^*}}{1 + e^{\nu_t^*}} \quad \text{with} \quad \nu_t^* = (1 - \rho_\nu)\bar{\nu} + \rho_\nu\nu_{t-1}^* + \epsilon_{\nu,t}$$

- Decisions to increase / decrease housing capital have **no impact** on liquidity (i.e. the probability of finding a trading partner)
- Understandable for tractability ...
- Related: simulated shock: **50% increase** in the avrg. months-to-sell... quite big? Specially compared to the shock fed into the SVAR

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Some minor notes

- **Cobb-Douglass between c and h:** seems to me it plays an important role in generating changes in output following a liquidity shock.
- **Calibration** How should I interpret the calibration of $\psi = 30\%$? Or how to interpret ψ altogether?
- Is this a paper intended to capture the crisis, or business cycles?